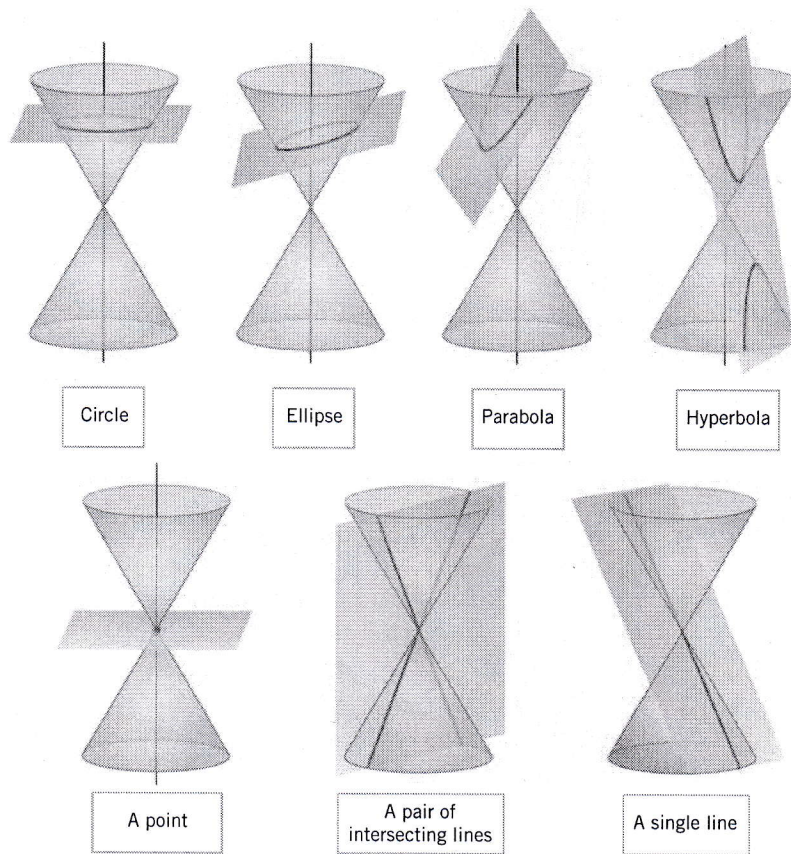


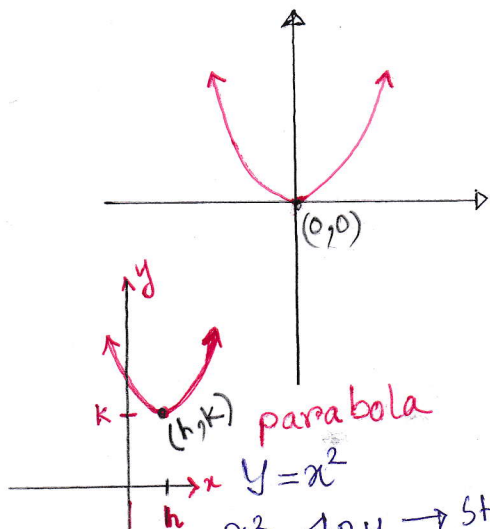
Conic section

■ CONIC SECTIONS

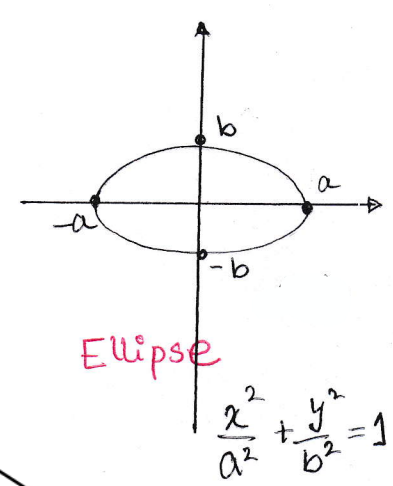
Circles, ellipses, parabolas, and hyperbolas are called *conic sections* or *conics* because they can be obtained as intersections of a plane with a double-napped circular cone (Figure 10.4.1). If the plane passes through the vertex of the double-napped cone, then the intersection is a point, a pair of intersecting lines, or a single line. These are called *degenerate conic sections*.



▲ Figure 10.4.1



parabola
 $y = x^2$
 $x^2 = 4py$ → Standard form
 Vertex: $(0,0)$ (turning point)
 $(x-h)^2 = 4p(y-k)$; $p > 0$ or $p < 0$
 Vertex: (h,k)

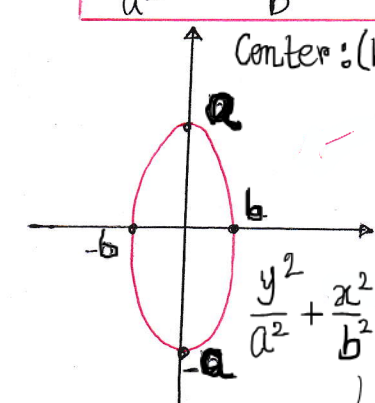


Ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \rightarrow \text{Center: } (0,0)$$

$$\frac{(y-k)^2}{a^2} + \frac{(x-h)^2}{b^2} = 1$$

Center: (h,k)

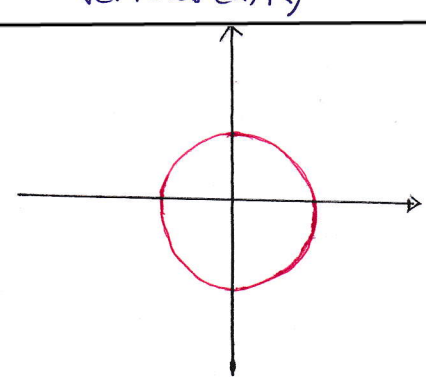


$$\frac{y^2}{a^2} + \frac{x^2}{b^2} = 1$$

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

→ Center: (h,k)

If it was not centered at the origin



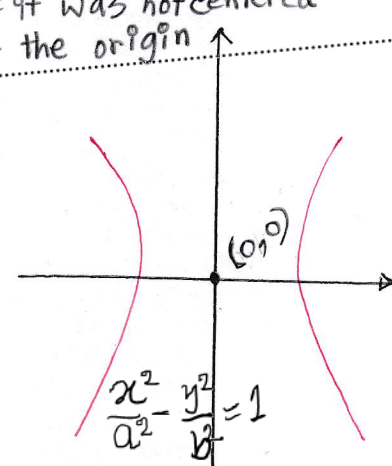
Circle

$$x^2 + y^2 = r^2$$

Center: $(0,0)$

$$(x-h)^2 + (y-k)^2 = r^2$$

Center: (h,k)

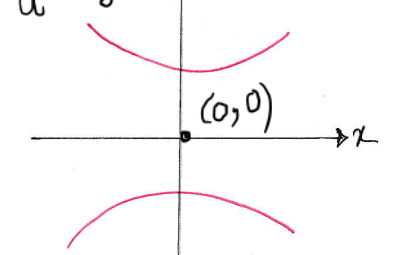


$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Hyperbola

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$



$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

→ Center (h,k)

General Equation of Second Degree

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0 \quad \text{--- (1)}$$

or $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

$$\underbrace{abc}_1 \quad \underbrace{fgh}_2$$

$$\begin{bmatrix} x & y & 1 \end{bmatrix} \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = 0$$

→ (2)

Relabeling the Constants

$$A=a; B=2h; C=b \\ D=2g; E=2f; F=c$$

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

We need to find the following invariants

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

discriminant of Eqn (1)

$$C = h^2 - ab$$

$$I = a + b$$

$$\Delta = 0$$

weak

$$\Delta \neq 0$$

strong

$$B^2 - 4AC \\ (2h)^2 - 4ab \\ 4h^2 - 4ab \\ 4(h^2 - ab) \\ (h^2 - ab)$$

↳ we can ignore the constant "4"

Invariants are some unchanged constants which indicates the locus of conic.

Locus — a set of points, the position of which satisfy a set of algebraic condition, such as circle, ellipse etc.

" Δ " identifies whether the conic is "Degenerate" or not (weak conic, quality of conic reduces)

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = a(bc - f^2) - h(hc - fg) + g(hf - bg) \\ = abc - af^2 - ch^2 + fgh + fgh - bg^2 \\ = abc + 2fgh - af^2 - bg^2 - ch^2$$

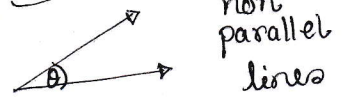
For degenerate case $\Delta = 0$; For proper conic $\Delta \neq 0$

Degenerate Case: $\Delta = 0$ $\rightarrow abc + 2fgh - af^2 - bg^2 - ch^2$

$$C = h^2 - ab, I = a + b$$

$\Delta = 0, C = 0$ If $h^2 - ab = 0$ then Parallel lines or straight line

$\Delta = 0, I = 0$ If $a + b = 0$ then perpendicular line
 $C \neq 0, I \neq 0$ If $\Delta = 0$ only but the above conditions do not fulfil then pair of st. lines.



Non-Degenerate Case: $\Delta \neq 0$

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

$C = 0 \Rightarrow h^2 - ab = 0 \rightarrow$ Parabola

$C > 0 \Rightarrow h^2 - ab > 0 \rightarrow$ hyperbola

$C < 0 \Rightarrow h^2 - ab < 0 \rightarrow$ Ellipse

$\left. \begin{matrix} h^2 - ab < 0 \\ a = b \\ h = 0 \end{matrix} \right\} \rightarrow$ Circle

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

What is represented by the following equation(s)?

① $4x^2 - 24xy - 6y^2 + 4x - 12y + 1 = 0$

$\underbrace{4x^2}_{2h} \quad \underbrace{-24xy}_b \quad \underbrace{-6y^2}_{2g} \quad \underbrace{+4x}_{2f} \quad \underbrace{-12y}_c \quad \underbrace{+1}_{c}$

$$\begin{aligned} 2h &= -24 \\ h &= -12 \end{aligned}$$

$$\begin{aligned} 2g &= 4 \\ g &= 2 \end{aligned}$$

$$\begin{aligned} 2f &= -12 \\ f &= -6 \end{aligned}$$

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

$$= (4)(-6)(1) + 2(-6)(2)(-12) - 4(-6)^2 - (-6)(2)^2 - (-1)(-12)^2$$

$$= 0$$

Degenerate case

$$\therefore C \neq 0, I \neq 0, \Delta = 0$$

$$\begin{aligned} C &= h^2 - ab = (-12)^2 - (4)(-6) = 168 \neq 0 \\ I &= a + b = 4 + (-6) = -2 \neq 0 \end{aligned}$$

The eqn represents a pair of st. lines

ii

$$x^2 - 4xy + y^2 + 8x + 2y - 5 = 0$$

$$a=1$$

$$2h = -4 \Rightarrow h = -2$$

$$b=1$$

$$c = -5$$

$$2g = 8$$

$$g = 4$$

$$2f = 2$$

$$f = 1$$

$$C = h^2 - ab = (-2)^2 - (1)(1) = 3 > 0$$

$$\therefore \Delta \neq 0, C > 0$$

The eqn represents a hyperbola

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

$$= -5 + (-16) - 1 - 16 - (-20)$$

$$= -18 \neq 0$$

iii

$$8x^2 + 4xy + 5y^2 - 16x - 14y + 13 = 0$$

$$a=8$$

$$b=5$$

$$c=13$$

$$2h = 4$$

$$h = 2$$

$$2g = -16$$

$$g = -8$$

$$2f = -14$$

$$f = -7$$

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

$$= 520 + 224 - 392 - 320 - 52$$

$$= -20 \neq 0$$

$$C = h^2 - ab = 2^2 - (8)(13) = -100 < 0$$

$\therefore \Delta \neq 0, C < 0$. The eqn represents an ellipse
 $a \neq b, h \neq 0$

iv

$$x^2 - 6xy + 9y^2 + 4x + 8y + 15 = 0$$

$$a=1$$

$$b=9$$

$$c=15$$

$$2h = -6$$

$$h = -3$$

$$2g = 4$$

$$g = 2$$

$$2f = 8$$

$$f = 4$$

$$h^2 - ab = (-3)^2 - (1)(9) = 0$$

$$C = h^2 - ab = 0$$

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

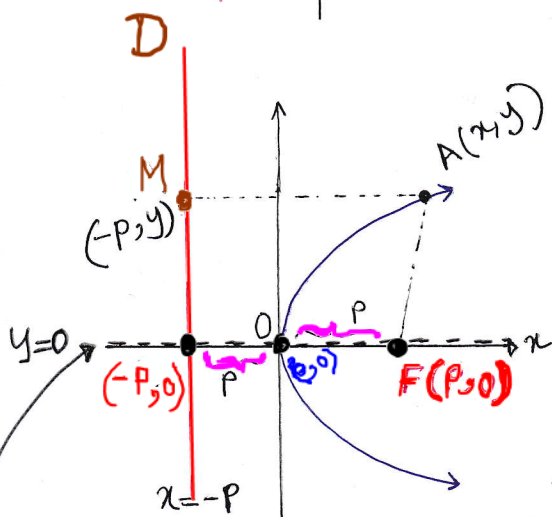
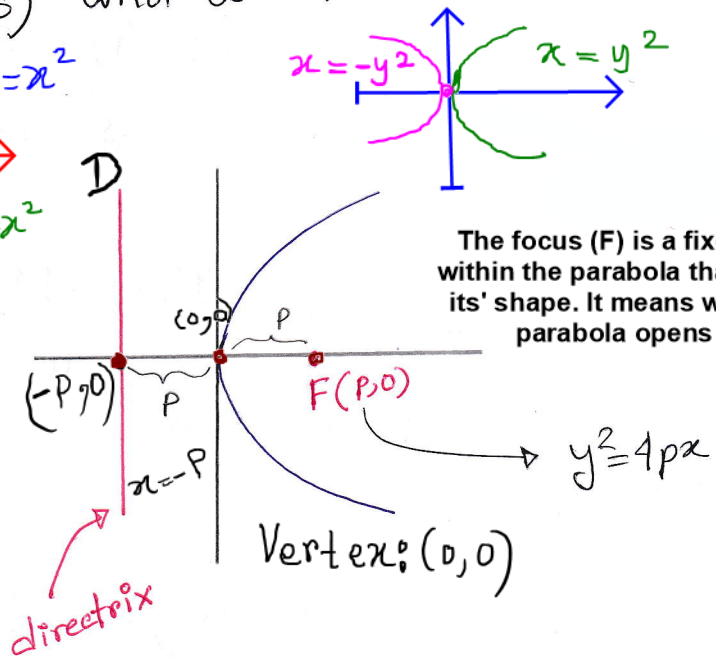
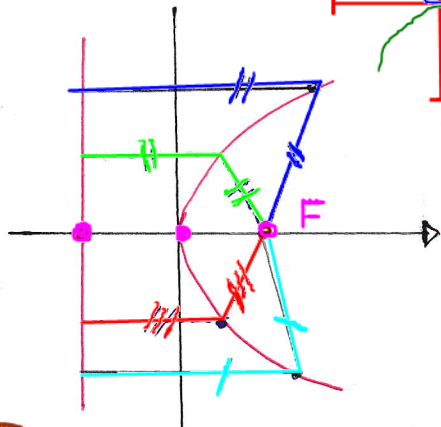
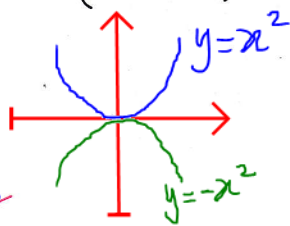
$$= (1)(9)(15) + 2(4)(2)(-3) - (1)(4)^2 - (9)(2)^2 - (15)(-3)^2$$

$$= -100$$

$\therefore \Delta \neq 0, C = 0$, Hence the eqn represents a parabola.

Parabola

Set of points in a plane that are equidistant from a fixed point F (focus) and a fixed line (directrix)



$$FA = AM$$

$$\sqrt{(x-p)^2 + (y-0)^2} = \sqrt{(x-(-p))^2 + (y-y)^2}$$

$$\sqrt{(x-p)^2 + y^2} = \sqrt{(x+p)^2}$$

$$\begin{aligned} (x-p)^2 + y^2 &= (x+p)^2 \\ x^2 - 2xp + p^2 + y^2 &= x^2 + 2xp + p^2 \end{aligned} \quad \left\{ \begin{array}{l} \text{square} \\ \text{both} \\ \text{sides} \end{array} \right.$$

$$y^2 = 4px$$

Vertex, $O = (0,0)$

Focus, $F = (p,0)$

Eqn of directrix, $D: x = -p$

$$(y-k)^2 = 4p(x-h)$$

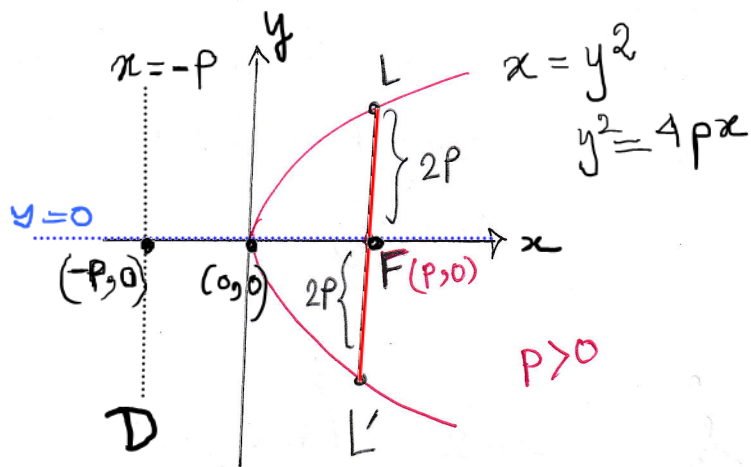
→ if the vertex shifts from the origin $(0,0)$

Axis of symmetry: the line that divides the parabola into two equal parts: $y = 0$

Recall the case of Circle:

Center $(0,0)$: $x^2 + y^2 = r^2$

Center (h,k) : $(x-h)^2 + (y-k)^2 = r^2$

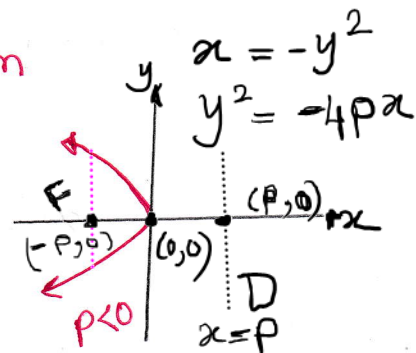


Draw a perpendicular line on Focus to create Latus Rectum LL'

LL' : Latus Rectum

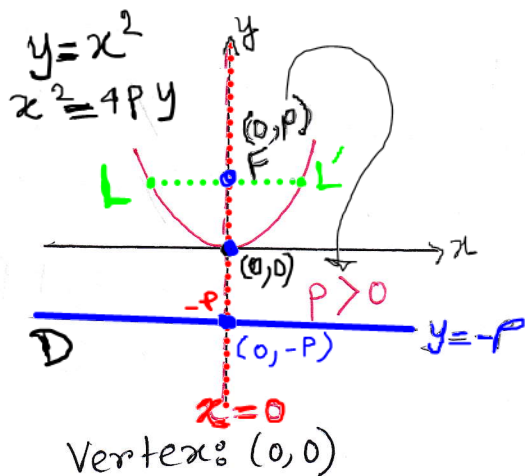
$$\therefore LL' = 4p$$

$$p > 0 \quad \text{or} \quad p < 0$$



Eqn of Latus Rectum: $x = p$

Length of Latus Rectum: $LL' = |4p|$ $\rightarrow$ length is always +ve



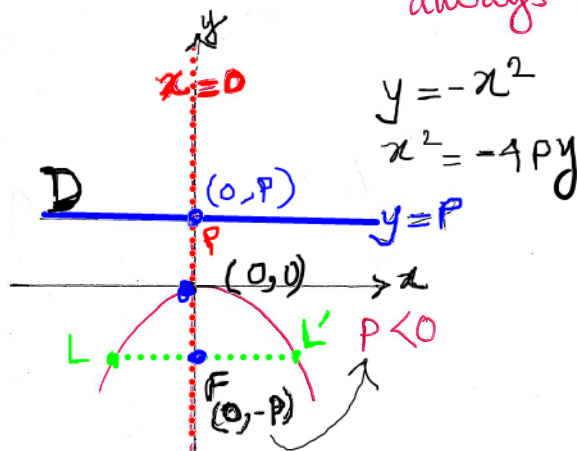
Vertex: $(0,0)$

Focus: $(0,p)$

(directrix) $D: y = -p$

axis of symmetry: $x = 0$

Latus Rectum, $LL' = |4p|$



Vertex: $(0,0)$

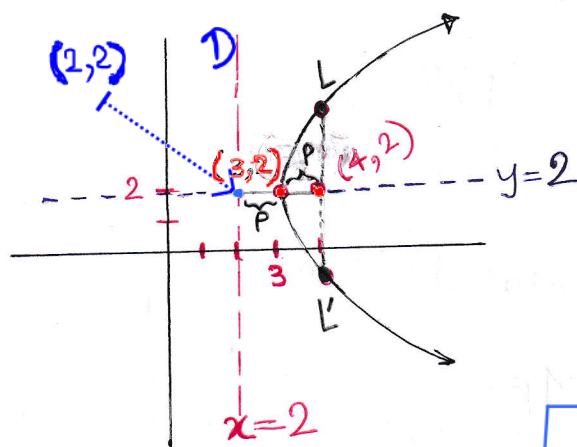
Focus: $(0,-p)$

$D: y = p$

axis of symmetry: $x = 0$

Latus Rectum, $LL' = |4p|$

Find the vertex, focus, equation of directrix, equation of axis of symmetry, length of Latus rectum of the parabola: $(y-2)^2 = 4(x-3)$



$$(h, k) = (3, 2)$$

$$4p = 4$$

$$p = 1$$

$$\text{Focus: } (p, 0)$$

$$= (4, 2)$$

$$X = p \quad Y = 0$$

$$x-3 = p \quad y-2 = 0$$

$$x-3 = 1 \quad y = 2$$

$$x = 4$$

$$(y-2)^2 = 4(x-3)$$

$$Y^2 = 4pX$$

$$4p = 4$$

$$p = 1$$

$$Y = 0 \Rightarrow y-2 = 0$$

$$y = 2$$

$$X = 0 \Rightarrow x-3 = 0$$

$$x = 3$$

$$\text{Vertex: } (3, 2)$$

$$\text{Vertex: } (3, 2)$$

$$\text{Focus: } (4, 2)$$

$$\text{Eqn of directrix: } x = 2$$

$$\text{Eqn of axis: } y = 0$$

$$y-2 = 0$$

$$y = 2$$

$$\text{Eqn of directrix: } x = -p$$

$$\Rightarrow x + p = 0$$

$$\Rightarrow (x-3) + p = 0$$

$$\Rightarrow x = 2$$

$$\begin{aligned} \text{Length of Latus Rectum: } |4p| \\ (LL') &= |4(1)| \\ &= 4 \end{aligned}$$

Find the vertex, focus, directrix, axis and length of latus rectum for the parabola, $y^2 - 2y - 4x - 3 = 0$

$$y^2 - 2y - 4x - 3 = 0 \rightarrow \text{General form}$$

$$y^2 - 2 \cdot y \cdot 1 + 1^2 - 1^2 - 4x - 3 = 0 \quad \text{completing square}$$

$$(y-1)^2 - 4x - 4 = 0$$

$$(y-1)^2 = 4(x+1)$$

$$(y-k)^2 = 4(x-h)$$

Standard form

$$(y-1)^2 = 4p(x+1)$$

$$y^2 = 4pX$$

$$4p = 4$$

$$\therefore \boxed{p = 1}$$

(i) Vertex: $(-1, 1) = (h, k)$

(ii) Focus: $(p, 0) = (0, 1)$

$$X = p$$

$$x+1 = 1$$

$$x = 0$$

$$Y = 0$$

$$y-1 = 0$$

$$y = 1$$

(iii) Eqn of directrix: $x = -2$

$$X = -p$$

$$x+1 = -1$$

$$x = -2$$

(v) Length of Latus Rectum: $|4p|$

$$= |4(1)|$$

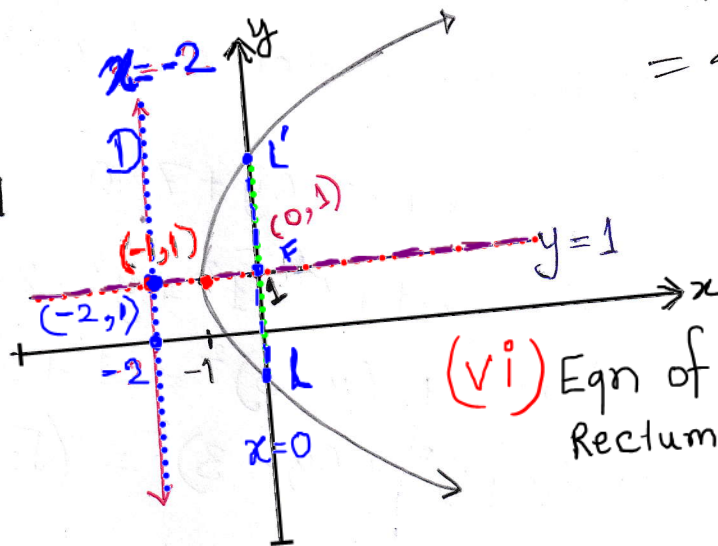
$$= 4$$

(iv) Axis of symmetry: $y = 1$

$$Y = 0$$

$$y-1 = 0$$

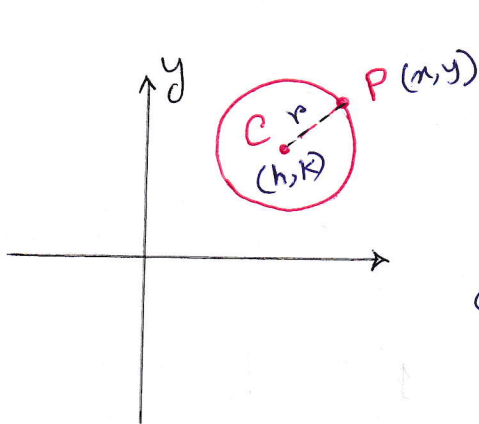
$$y = 1$$



(vi) Eqn of Latus Rectum: $x = 0$

Circle

Set of points in a plane that are equidistant from a fixed point C (center).

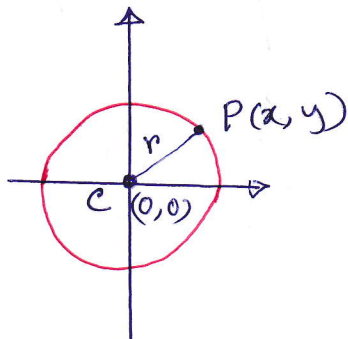


$$|PC| = \sqrt{(x-h)^2 + (y-k)^2} = r$$

$$\Rightarrow (x-h)^2 + (y-k)^2 = r^2$$

center: (h, k)

↓
standard eqn of the circle. Square both sides

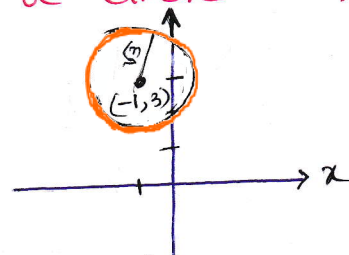


$$|PC| = \sqrt{(x-0)^2 + (y-0)^2} = r$$

$$\Rightarrow x^2 + y^2 = r^2$$

Center: (0, 0)

Question: Sketch the graph with the equation $x^2 + y^2 + 2x - 6y + 7 = 0$ by first showing that it represents a circle and then finding its center and radius.



$$x^2 + 2x + y^2 - 6y + 7 = 0$$

$$\Rightarrow x^2 + 2 \cdot x \cdot 1 + 1^2 - 1^2 + y^2 - 2 \cdot y \cdot 3 + 3^2 - 3^2 + 7 = 0$$

$$\Rightarrow (x+1)^2 + (y-3)^2 = 1^2 + 3^2 - 7 = 3$$

$$\Rightarrow (x+1)^2 + (y-3)^2 = (\sqrt{3})^2$$

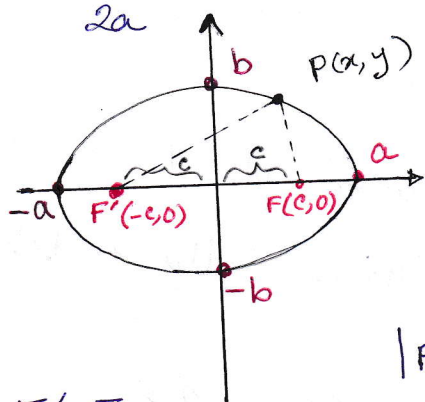
Center: (-1, 3)

radius: $\sqrt{3}$

Ellipse

An ellipse is the set of all points in the plane the sum of whose distances from two fixed points (the foci) is a constant.

foci $\rightarrow$ plural of focus



$$a > b$$

$$|PF'| + |PF| = 2a$$

$\underbrace{F', F}_{\text{foci}}$

$$\Rightarrow \sqrt{(x+c)^2 + y^2} + \sqrt{(x-c)^2 + y^2} = 2a$$

$$\Rightarrow \sqrt{(x-c)^2 + y^2} = 2a - \sqrt{(x+c)^2 + y^2}$$

$$\Rightarrow \cancel{x^2} - 2cx + \cancel{c^2} + y^2 = 4a^2 - 4a\sqrt{(x+c)^2 + y^2} + \cancel{x^2} + 2cx + \cancel{c^2} + y^2$$

(sq both sides)

$$\Rightarrow a\sqrt{(x+c)^2 + y^2} = a^2 + cx$$

$$\Rightarrow a^2(x^2 + 2cx + c^2 + y^2) = a^4 + 2a^2cx + c^2x^2$$

(sq both sides)

$$\Rightarrow a^2x^2 + \cancel{2a^2cx} + a^2c^2 + a^2y^2 = a^4 + \cancel{2a^2cx} + c^2x^2$$

$$\Rightarrow a^2x^2 - c^2x^2 + a^2y^2 = a^4 - a^2c^2$$

$$\Rightarrow (a^2 - c^2)x^2 + a^2y^2 = a^2(a^2 - c^2)$$

$$\Rightarrow b^2x^2 + a^2y^2 = a^2b^2$$

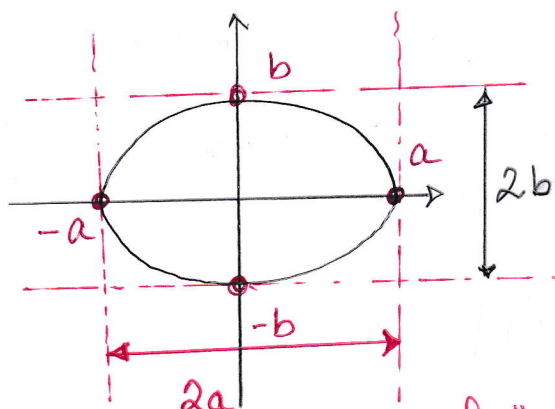
Redefine the constant
 $a^2 - c^2 = b^2$

$$\Rightarrow \frac{b^2 x^2}{a^2 b^2} + \frac{a^2 y^2}{a^2 b^2} = \frac{a^2 b^2}{a^2 b^2} \quad (\div \text{ by } a^2 b^2)$$

$$\Rightarrow \boxed{\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1} \rightarrow \text{Eqn of Ellipse}$$

Center (0,0)

$a > b$



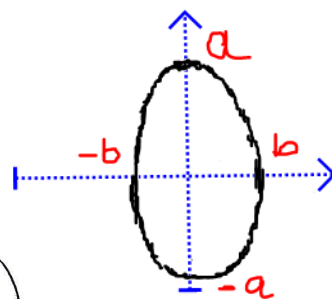
$2b \rightarrow$ distance of minor axis $\because a > b$

(Horizontally stretched)

$2a \rightarrow$ distance of the major axis. $\because a > b$

$$\boxed{\frac{y^2}{a^2} + \frac{x^2}{b^2} = 1}$$

(Vertically stretched)



Ratio between the length of the focus and the length of the semi-major axis (one of the vertices)

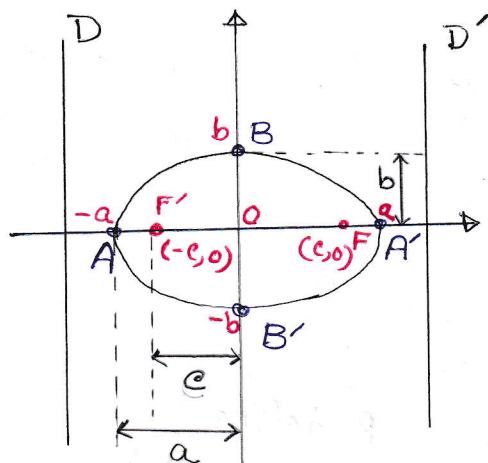
$$\text{Eccentricity, } e = \frac{c}{a}$$

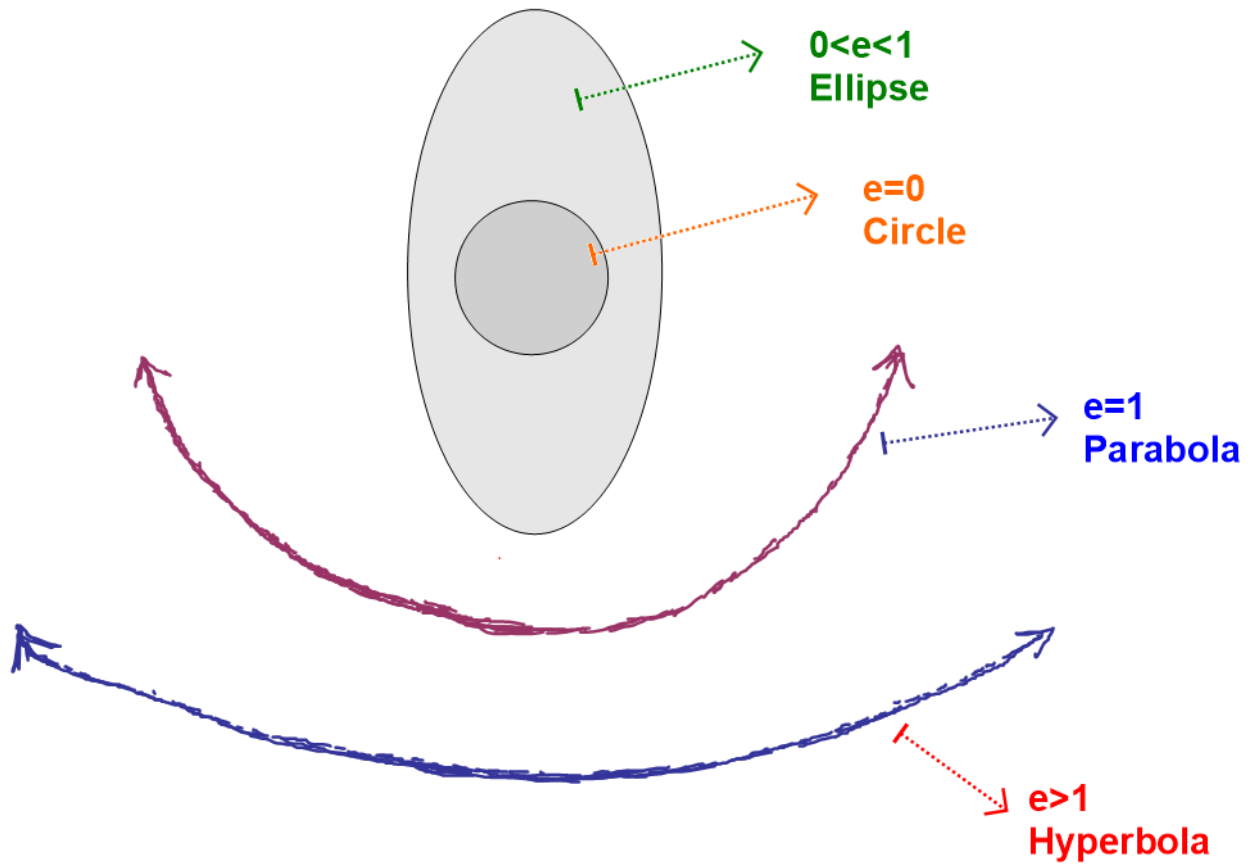
$\downarrow$
It is a constant

{ It defines how circular or uncircular the shape of the conic is }

$$0 < e < 1$$

$$\boxed{c < a}$$





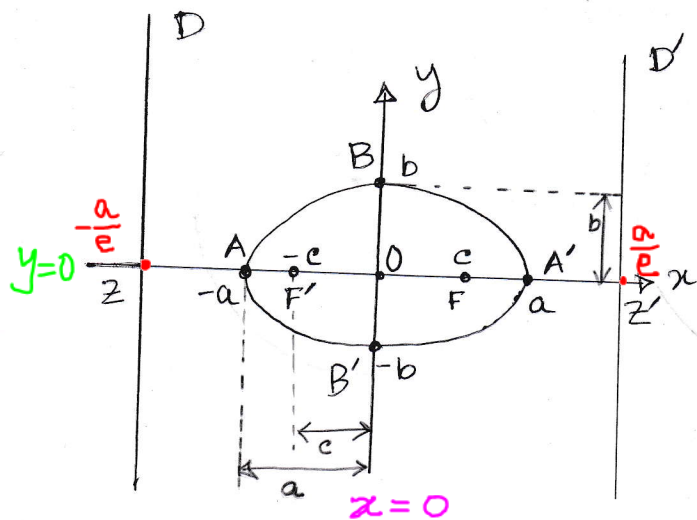
$$e = \frac{c}{a}$$

$$e = \frac{\sqrt{a^2 - b^2}}{a}$$

$a^2 - c^2 = b^2 \quad \because$ we redefined this constant

$$\Rightarrow c^2 = a^2 - b^2$$

$$\Rightarrow c = \sqrt{a^2 - b^2}$$



Centre, $O : (0,0)$
 Foci, $(\pm c, 0)$ or $(\pm ae, 0)$
 $\rightarrow F' \& F$

$$\begin{cases} e = \frac{c}{a} \\ c = ae \end{cases}$$

Vertices $(\pm a, 0) \rightarrow A \& A'$

Covertices $(0, \pm b) \rightarrow B \& B'$

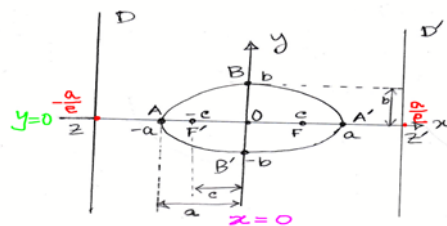
Equation of major axis: $y = 0 \rightarrow AA'$ (the line)

Equation of minor axis: $x = 0 \rightarrow BB'$ (the line)

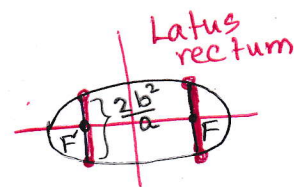
Length of major axis: $2a$ (AA')

Length of minor axis: $2b$ (BB')

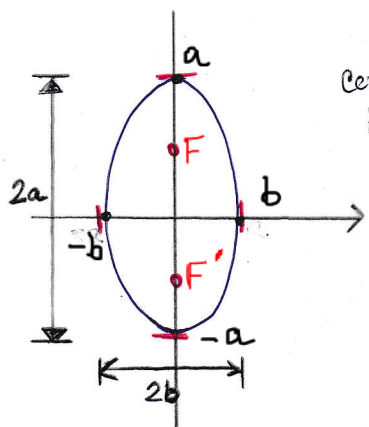
Equation of directrices: $DZ \& D'Z'$



Equation of latus rectum: $x = \pm \frac{a}{e}$
 $\begin{cases} x = \pm c \\ x = \pm ae \end{cases}$
 $\because c = ae$



Length of Latus rectum: $\frac{2b^2}{a}$



center: $(0,0)$

Foci: $(0, \pm ae)$

Vertices: $(0, \pm a) \rightarrow$ major axis

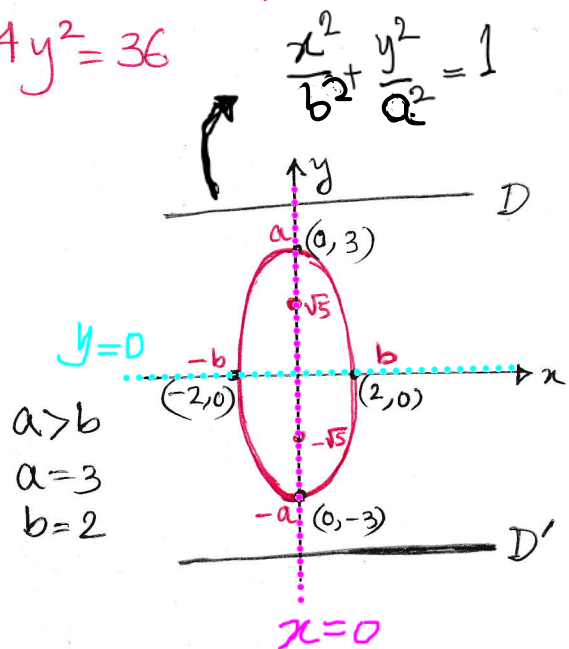
Covertices: $(\pm b, 0) \rightarrow$ minor axis

Example: Find all the characteristics of the graph with equation: $9x^2 + 4y^2 = 36$

$$\frac{x^2}{4} + \frac{y^2}{9} = 1 \quad (\div \text{ by } 36)$$

$$\Rightarrow \frac{x^2}{2^2} + \frac{y^2}{3^2} = 1$$

$$\rightarrow \frac{y^2}{3^2} + \frac{x^2}{2^2} = 1$$



Center: $(0, 0)$

Vertices: $(0, \pm a) = (0, \pm 3) \rightarrow$ major axis

Co-vertices: $(\pm b, 0) = (\pm 2, 0) \rightarrow$ minor axis

$$e = \frac{c}{a} \quad \begin{aligned} a^2 - c^2 &= b^2 & c^2 &= 3^2 - 2^2 = 5 & \therefore c &= \frac{c}{a} \\ a^2 - b^2 &= c^2 & c &= \pm \sqrt{5} & &= \frac{\sqrt{5}}{3} \end{aligned}$$

Foci: $(0, \pm c) = (0, \pm \sqrt{5})$

Equation of major axis: $x=0$

Eqn of minor axis: $y=0$

Length of major axis: $2a = 2(3) = 6$

Length of minor axis: $2b = 4$

Equation of directrices: $y = \pm \frac{a}{e} = \pm \frac{3}{\sqrt{5}/3} = \pm \frac{9}{\sqrt{5}}$

Eqn of latus rectum: $y = \pm c = \pm \sqrt{5}$

Length of latus rectum: $\frac{2b^2}{a} = \frac{8}{3}$

Compare

$$\frac{(y-k)^2}{a^2} + \frac{(x-h)^2}{b^2} = 1$$

$$\frac{y^2}{a^2} + \frac{x^2}{b^2} = 1$$

if center is not
at the origin;
Center: (h, k)

Vertically stretched

Standard forms of the eqn of an ellipse with center: (h, k)

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1 \quad \leftarrow \text{Horizontally Stretched}$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

• length of major axis = $2a$

• " " minor " " = $2b$

$$(\pm a, 0) = (X, Y)$$

$$\begin{aligned} X &= \pm a & Y &= 0 \\ x-h &= \pm a & y-k &= 0 \\ x &= h \pm a & y &= k \end{aligned}$$

• coordinates of the vertices: $(h \pm a, k)$

• coordinates of the covertices: $(h, k \pm b)$

• coordinate of the foci: $(h \pm c, k)$

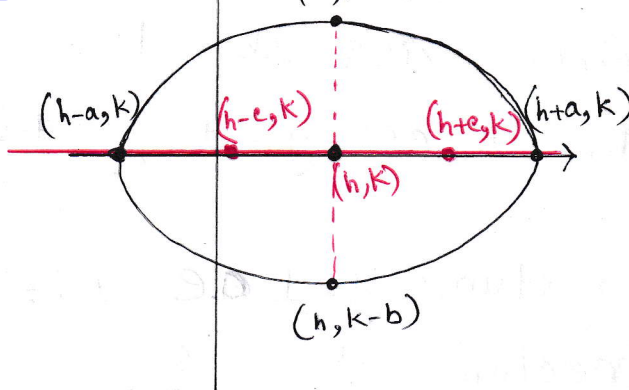
$$(0, \pm b) = (X, Y)$$

$$\begin{aligned} X &= 0 & Y &= \pm b \\ x-h &= 0 & y-k &= \pm b \\ x &= h & y &= k \pm b \end{aligned}$$

$$(X, Y) = (\pm c, 0)$$

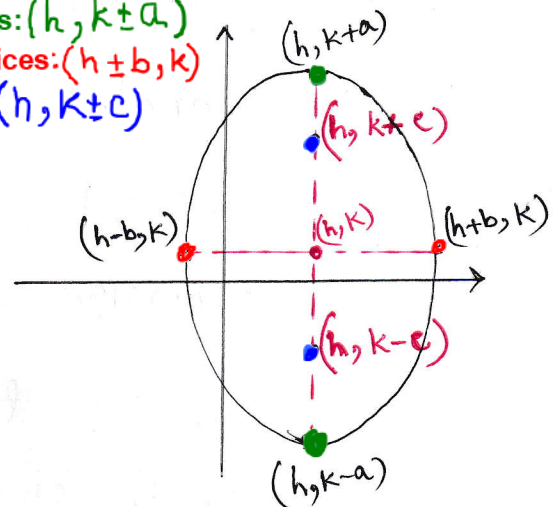
$$\begin{aligned} X &= \pm c & Y &= 0 \\ x-h &= \pm c & y-k &= 0 \\ x &= h \pm c & y &= k \end{aligned}$$

Vertices: $(h \pm a, k)$
Covertices: $(h, k \pm b)$
Foci: $(h \pm c, k)$



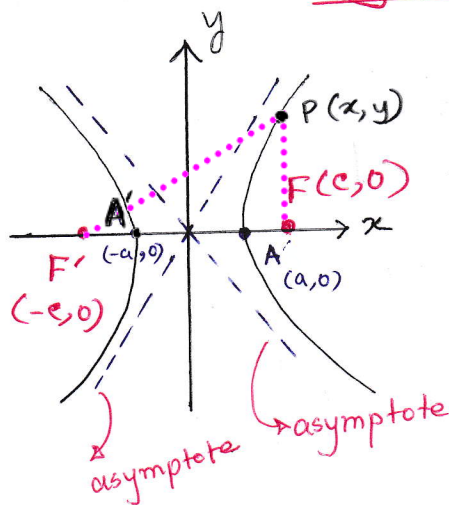
Horizontally stretched

Vertices: $(h, k \pm a)$
Covertices: $(h \pm b, k)$
Foci: $(h, k \pm c)$



Vertically stretched

Hyperbola



Set of all points in a plane the difference of whose distances from two fixed points (the foci) is a constant.

$F \text{ \& } F'$

$$|PF| - |PF'| = \pm 2a$$

$$PF > PF' = 2a$$

$$PF < PF' = -2a$$

$$\Rightarrow \sqrt{(x-c)^2 + y^2} - \sqrt{(x+c)^2 + y^2} = \pm 2a$$

$$\Rightarrow \sqrt{(x-c)^2 + y^2} = \pm 2a + \sqrt{(x+c)^2 + y^2}$$

$$\Rightarrow \cancel{x^2} - 2cx + \cancel{c^2} + y^2 = 4a^2 \pm 2a\sqrt{(x+c)^2 + y^2} + \cancel{x^2} + 2cx + \cancel{c^2} + y^2$$

$$\Rightarrow -4cx = 4a^2 \pm 4a\sqrt{(x+c)^2 + y^2}$$

$$\Rightarrow \pm a\sqrt{(x+c)^2 + y^2} = a^2 + cx \quad (\div \text{ by } 4)$$

$$\Rightarrow a^2(x^2 + 2cx + c^2 + y^2) = a^4 + 2a^2cx + c^2x^2$$

$$\Rightarrow a^2x^2 + 2a^2cx + a^2c^2 + y^2 \cancel{a^2} = a^4 + 2a^2cx + c^2x^2$$

$$\Rightarrow a^2x^2 + a^2c^2 + y^2 \cancel{a^2} = a^4 + c^2x^2$$

$$\Rightarrow a^2x^2 + a^2(a^2 + b^2) + a^2y^2 = a^4 + (a^2 + b^2)x^2$$

Redefine the constant

$$\Rightarrow \cancel{a^2x^2} + \cancel{a^4} + a^2b^2 + a^2y^2 = \cancel{a^4} + \cancel{a^2x^2} + b^2x^2$$

$$\Rightarrow a^2b^2 = b^2x^2 - a^2y^2$$

$$\Rightarrow \boxed{\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1}$$

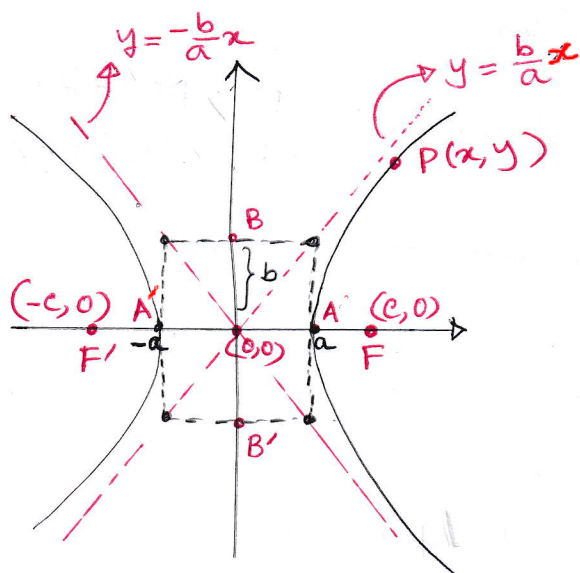
17

($\div$ by a^2b^2) In ellipse we redefine $c^2 = a^2 - b^2$

standard form

Center: $(0,0) \Rightarrow \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Center: $(h,k) \Rightarrow \frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$



Center: $C(0,0)$

Vertices: $A \text{ \& } A'$

$$\Rightarrow (\pm a, 0)$$

Foci: $F \text{ \& } F'$

$$\Rightarrow (\pm c, 0)$$

$$e = \frac{c}{a}$$

$$e > 1$$

See graph at pg 13

Eccentricity: $e = \frac{c}{a}$ while $c = \sqrt{a^2 + b^2}$

This sign is opposite to ellipse

Eqn of Asymptotes: $y = \pm \left(\frac{b}{a}\right)x$

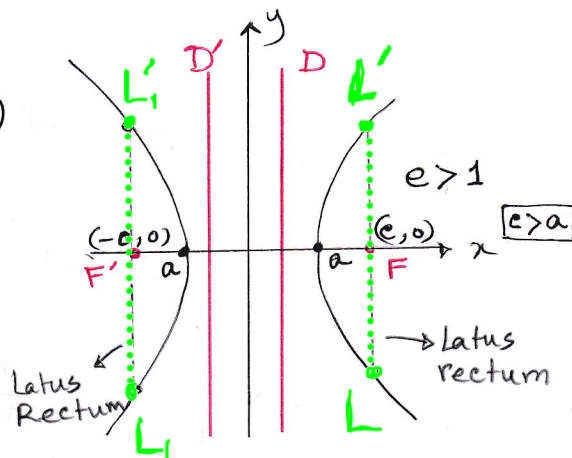
Length of transverse axis: $2a$ (AA')

Length of conjugate axis: $2b$ (BB')

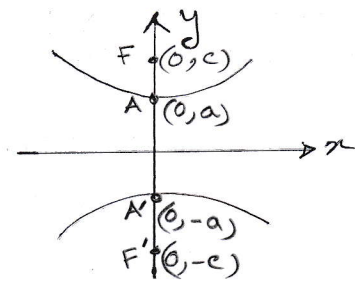
Eqn of directrices: $x = \pm \frac{a}{e}$
($D \text{ \& } D'$)

Length of Latus rectum: $\frac{2b^2}{a}$

Eqn of Latus rectum:
 $x = -c, x = c$
Foci



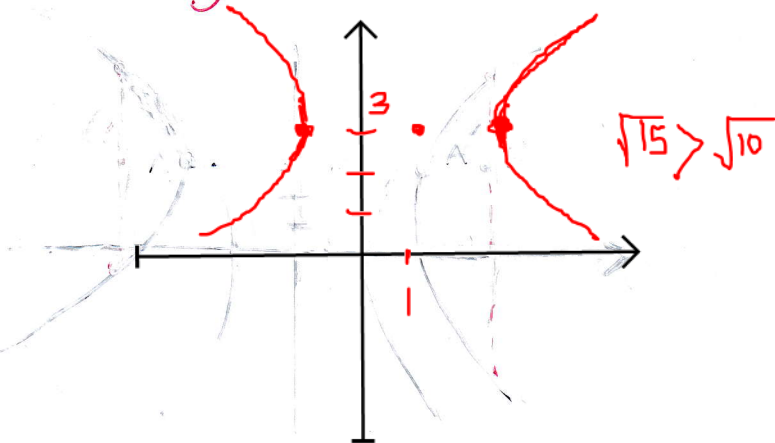
If we have: $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$



Vertices: $(0, \pm a)$

Foci: $(0, \pm c)$

Example: Find all the characteristics of the hyperbola with equation $2(x-1)^2 - 3(y-3)^2 = 30$



$$2(x-1)^2 - 3(y-3)^2 = 30$$

$$\frac{(x-1)^2}{15} - \frac{(y-3)^2}{10} = 1$$

$$a = \sqrt{15}, \quad b = \sqrt{10}$$

$$\text{Center, } C = (1, 3)$$

$$\text{Vertices} = (1 \pm \sqrt{15}, 3) \quad \frac{X^2}{a^2} - \frac{Y^2}{b^2} = 1 \quad \text{Vertices: } (\pm a, 0)$$

$$c = \sqrt{a^2 + b^2} = \sqrt{15 + 10} = 5$$

$$e = \frac{c}{a} = \frac{5}{\sqrt{15}}$$

$$\begin{aligned} X &= \pm a \\ \Rightarrow x-1 &= \pm a \Rightarrow y-3=0 \\ \Rightarrow x &= 1 \pm a \Rightarrow y=3 \end{aligned}$$

$$x = 1 \pm \sqrt{15}$$

$$(X, Y)$$

$$\text{Foci: } (\pm c, 0)$$

$$X = \pm c = \pm 5$$

$$x-1 = \pm 5$$

$$x = 1 \pm 5 = 6, -4$$

$$\begin{aligned} &\left\{ \begin{array}{l} Y=0 \\ y-3=0 \\ y=3 \end{array} \right\} \text{Foci: } (6, 3) \text{ and } (-4, 3) \\ &\quad (X, Y) \end{aligned}$$

Eqn of directrices: $X = \pm \frac{a}{e}$

$$x-1 = \pm \frac{\sqrt{15}}{5/\sqrt{15}} = \pm 3$$

$$x = 1 \pm 3$$

$$x = 4, x = -2$$

Eqn of asymptotes: $y = \pm \frac{b}{a}x$

$$y-3 = \pm \frac{\sqrt{10}}{\sqrt{15}}(x-1)$$

$$y = 3 \pm \frac{\sqrt{2}}{\sqrt{3}}(x-1)$$

$$= 3 \pm \frac{\sqrt{6}}{3}(x-1)$$

$$y = \frac{\sqrt{6}}{3}(x-1)+3 ; y = -\frac{\sqrt{6}}{3}(x-1)+3$$

Eqn of latus rectum:

$$x=6 \text{ and } x=-4$$

Foci

$$\begin{aligned} \text{Length of Latus rectum: } \frac{2b^2}{a} &= \frac{2(\sqrt{10})^2}{\sqrt{15}} \\ &= \frac{20}{\sqrt{15}} \end{aligned}$$

$$\begin{aligned} &= \frac{20\sqrt{15}}{15} \\ &= \frac{4\sqrt{15}}{3} \end{aligned}$$

Length of transverse axis: $2a = 2\sqrt{15}$
" " conjugate axis: $2b = 2\sqrt{10}$

